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GEN – AI

Question 1: Tokenization

(a) Split "INTERNATIONALIZATION" into 3 reasonable sub-word pieces.

INTER + NATIONAL + IZATION

Justification:

- INTER means "between" or "among."
- NATIONAL relates to a nation or country.
- IZATION is a suffix used to indicate a process or transformation.

Together, these sub-words combine to form the meaning of making something international.

(b) Why does sub-word tokenization help instead of whole-word tokenization?

Sub-word tokenization helps because many words are rare, long, or may not appear in the training data. By breaking a word into smaller meaningful units, the model can understand and process unfamiliar words using known sub-word components. This reduces the vocabulary size and allows the model to handle new words efficiently. For example, even if the model has never seen the complete word "INTERNATIONALIZATION," it may have learned the meanings of "INTER," "NATIONAL," and "IZATION," enabling it to infer the overall meaning of the word.

Question 2: Embeddings

(a) Which pair would have higher cosine similarity and why?

The pair **"king" and "queen"** would have a higher cosine similarity than **"king" and "apple."**

This is because "king" and "queen" belong to the same semantic category of royalty and share many contextual similarities in language. They often appear in similar sentences and discussions related to kingdoms, leadership, and monarchy. In contrast, "apple" is a fruit and has a completely different meaning and usage context. Therefore, the embedding vectors of "king" and "queen" are expected to be much closer together in the embedding space.

(b) What does an embedding represent, and how is it different from a one-hot vector?

An embedding is a dense numerical representation of a word that captures its meaning, context, and relationships with other words. Words with similar meanings tend to have similar embedding vectors.

A one-hot vector, on the other hand, is a sparse representation where only one position is set to 1 and all others are 0. It simply identifies a word without providing any information about its meaning or relationship to other words.

For example, a one-hot vector treats "king" and "queen" as completely unrelated symbols, while embeddings place them close together because they are semantically similar. Thus,

embeddings are much more useful for machine learning and natural language processing tasks.

Question 3: Self-Attention Worked Example (5 Marks)

Sentence: **"The city councilmen refused the demonstrators a permit because they feared violence."**

(a) "They" should refer to which word?

Answer: Councilmen

The word "they" refers to the **city councilmen** because it is more logical that the councilmen feared violence and therefore refused the permit.

(b) Why is self-attention needed to correctly resolve what "they" refers to?

Self-attention enables the model to examine the relationship between every word in a sentence, regardless of the distance between them. Instead of relying only on word order, it considers the context and meaning of the entire sentence.

In this example, self-attention helps determine that "they" is more closely related to "councilmen" because the action of refusing a permit due to fear of violence is more logically associated with the councilmen. By analyzing contextual relationships, self-attention allows the model to correctly resolve pronoun references and improve language understanding.

Question 4: Query, Key, Value (5 Marks)

Explain what Query, Key, and Value vectors represent in self-attention and how they produce an attention score.

In the self-attention mechanism, each word is transformed into three vectors known as **Query (Q)**, **Key (K)**, and **Value (V)**.

- **Query (Q):** Represents the information that a word is searching for from other words in the sentence.
- **Key (K):** Represents the characteristics or features of a word that can be matched with a query.
- **Value (V):** Contains the actual information that will be used in the final output.

The attention process works by comparing the Query vector of one word with the Key vectors of all other words. These comparisons generate attention scores that indicate how important each word is to the current word. The scores are then converted into weights using the Softmax function.

Finally, the Value vectors are multiplied by these attention weights and combined to produce a context-aware representation of the word. This allows the model to focus on the most relevant words in a sentence and capture important relationships between them.

Example: When processing the word "they," the Query vector can identify that "councilmen" is highly relevant. As a result, more attention is given to the Value vector of "councilmen," helping the model understand the correct meaning of "they."

